

Practice Paper B — Applied Core

100 marks total. These questions test judgement — diagnosing, deciding, and explaining — rather than calculation. Full marks require a reasoned explanation, not just a conclusion. Full worked solutions follow.

Question 1 (12 marks). A colleague proposes simplifying the recalibration engine by using a single volatility quote — taken from the ATM 5-year cap — as the input to every caplet across the entire curve, on the grounds that “it’s roughly the right level and saves a whole stripping step.” Explain, referencing specific chapters, why this is a bad idea, and describe what is lost.

Question 2 (14 marks). A SABR fit to a 5y × 10y swaption smile returns $\rho = -0.55$, $\nu = 0.15$, compared to yesterday’s fit of $\rho = -0.30$, $\nu = 0.38$ on the same underlying, with total fit residual essentially unchanged between the two days. Is this pair of parameter moves, taken together, more likely to reflect a genuine, large one-day shift in the market’s smile, or a calibration degeneracy? Justify your answer and describe one concrete check you would run before accepting the new numbers.

Question 3 (14 marks). A cascade calibration run produces excellent fits (sub-1 bp residual) for every co-terminal swaption with expiry beyond 7 years, and steadily worsening residuals (rising to 15 bp) for expiries inside 3 years. Diagnose the likely cause and recommend a specific remedy, referencing the relevant chapter.

Question 4 (12 marks). A junior quant implements a BGM Monte Carlo pricer for a Bermudan swaption using long-stepping (one time step per 3-month tenor period) with a full predictor-corrector scheme, and separately reuses the exact same simulation engine, unmodified, to price a strip of in-arrears SOFR caplets. The Bermudan price looks sensible; the in-arrears caplet prices are all noticeably too high relative to the market. Explain the likely cause.

Question 5 (12 marks). A trader asks why the desk bothers with a full joint, regularized calibration every morning when yesterday’s cascade-only calibration “looked basically fine.” Give two distinct, concrete reasons a cascade-only calibration is a materially weaker choice for a production system, referencing specific failure modes from the relevant chapter.

Question 6 (16 marks). The morning recalibration flags a validation failure on exactly one instrument: a 3y × 7y receiver swaption reprices 22 bp of normal vol away from its input quote, while every cap, and every other swaption on the grid, reprices within a 2 bp tolerance. Design a structured triage process (list concrete steps in order) to determine, within a tight pre-open time window, whether this is a data problem or a genuine market/model issue, and state what action you would take if the check is inconclusive by market open.

Question 7 (10 marks). Explain, to a risk manager with no quant background, in no more than five sentences, why a rank-3 correlation reduction is generally a reasonable simplification for a BGM model used for vanilla and moderately exotic products, but why it might not be adequate for a product whose value depends heavily on the precise co-movement of two very specific, nearby points on the curve.

Question 8 (10 marks). A newly onboarded currency’s options market has genuinely thin liquidity: only three swaption tenors and two cap maturities are reliably quoted, far fewer than the model’s full tenor grid. Describe, briefly, what specifically breaks or degrades in

the calibration pipeline of Chapters 16–18 under this data scarcity, and suggest one practical mitigation.